Practice Problem Set

Taylor Series · Maxima–Minima & the Hessian · Backpropagation & Automatic Differentiation

Part A — Taylor Series

Problem A1 · Taylor polynomial about a point

Let $f(x) = \ln x$. Compute f and its first four derivatives at $x = 1$, and write the fourth-order Taylor polynomial $P_4(x)$ of f about the point $a = 1$.

Solution

Differentiate and evaluate at the centre $a = 1$:

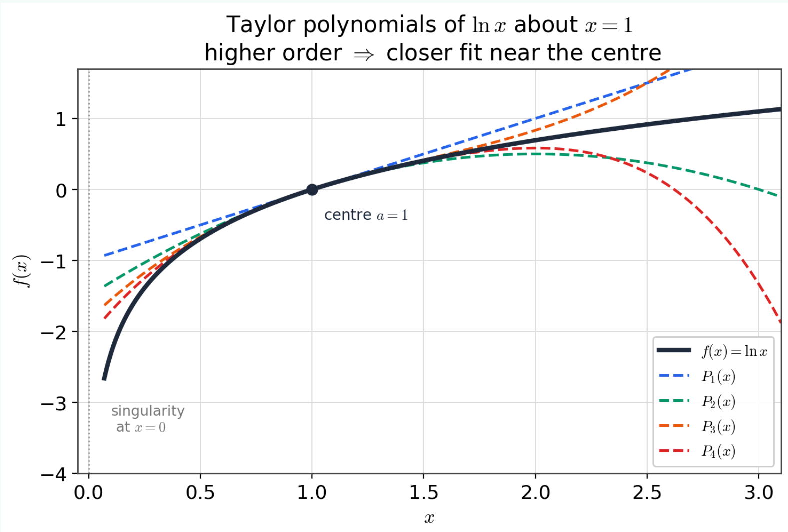
$$\begin{aligned} &f = \ln x, \quad f' = x^{-1}, \quad f'' = -x^{-2}, \quad f''' = 2x^{-3}, \quad f^{(4)} = -6x^{-4}, \\ &f(1) = 0, \quad f'(1) = 1, \quad f''(1) = -1, \quad f'''(1) = 2, \quad f^{(4)}(1) = -6. \end{aligned}$$

The fourth-order Taylor polynomial about $a = 1$ is

$$P_4(x) = f(1) + f'(1)(x-1) + \frac{f''(1)}{2!}(x-1)^2 + \frac{f'''(1)}{3!}(x-1)^3 + \frac{f^{(4)}(1)}{4!}(x-1)^4.$$

Substituting the values (and $\frac{2}{3!} = \frac{1}{3}$, $\frac{-6}{4!} = -\frac{1}{4}$),

$$\boxed{P_4(x) = (x-1) - \tfrac{1}{2}(x-1)^2 + \tfrac{1}{3}(x-1)^3 - \tfrac{1}{4}(x-1)^4}.$$



The Taylor polynomials $P_1, \ldots, P_4$ of $\ln x$ about $x = 1$: each added term tightens the fit near the centre $x = 1$.

Alternate / quicker solution

Substitute $u = x-1$ to reuse a series you already know: $\ln x = \ln(1+u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \frac{u^4}{4} + \cdots$. Truncating at degree 4 and putting $u = x - 1$ back gives the same $P_4(x)$ in a single line — no repeated differentiation needed.

Intuition — why it matters

A Taylor polynomial replaces f near a point a by the polynomial that matches f's value and first few derivatives there; the coefficient of $(x-a)^k$ is the scaled derivative $f^{(k)}(a)/k!$. The more terms kept (and the closer x is to the centre), the better the fit — the basis for the local quadratic models used later in optimisation.

Problem A2 · Second-order Taylor expansion in two variables

Let $f(x,y) = e^x \sin y$.

(a) Compute ∇f and the Hessian H at $(0,0)$.

(b) Write the second-order Taylor polynomial $Q(x,y)$ of f about $(0,0)$.

(c) Use Q to approximate $f(0.1, 0.2)$, compare with the exact value, and comment on the error using the figure.

Solution

The second-order expansion about (a,b) is

$$f(a+h, b+k) \approx f + \big(f_x h + f_y k\big) + \tfrac{1}{2}\big(f_{xx}h^2 + 2f_{xy}hk + f_{yy}k^2\big),$$

all derivatives evaluated at (a,b).

(a) The partials of $f = e^x \sin y$ are

$$f_x = e^x \sin y, \quad f_y = e^x \cos y, \quad f_{xx} = e^x \sin y, \quad f_{xy} = e^x \cos y, \quad f_{yy} = -e^x \sin y.$$

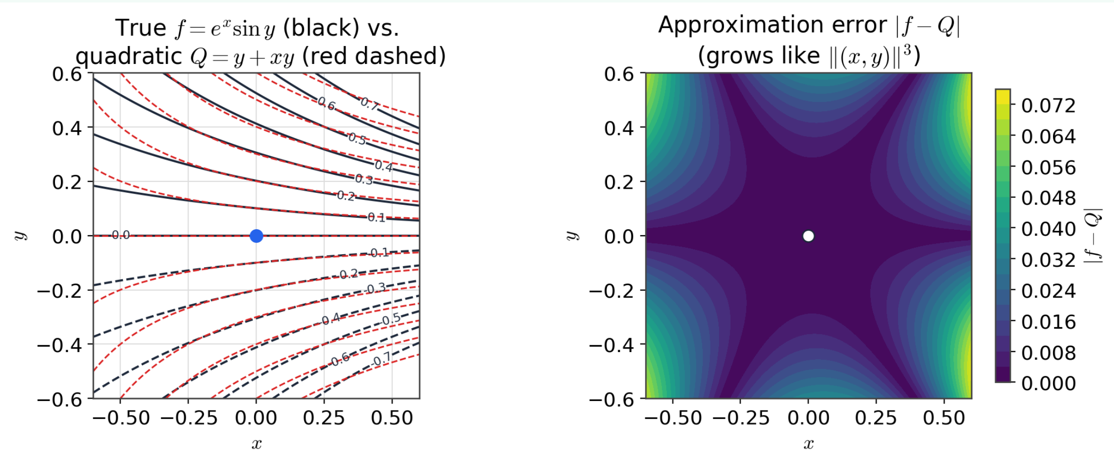
At $(0,0)$: $f=0$, $f_x = 0$, $f_y = 1$, $f_{xx} = 0$, $f_{xy} = 1$, $f_{yy} = 0$, so

$$\nabla f(0,0) = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad H(0,0) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

(b) With $(a,b) = (0,0)$, $h = x$, $k = y$:

$$Q(x,y) = 0 + (0\cdot x + 1\cdot y) + \tfrac{1}{2}\big(0\cdot x^2 + 2\cdot 1\cdot xy + 0\cdot y^2\big) = \boxed{y + xy}.$$

(c) $Q(0.1, 0.2) = 0.2 + (0.1)(0.2) = 0.22$. Exactly, $f(0.1, 0.2) = e^{0.1}\sin(0.2) = 1.10517 \times 0.19867 = 0.21956$, so the error is $+4.4\times10^{-4}$ (Q slightly overestimates). The first neglected term is third order ($O(\|(x,y)\|^3)$), which is why the error in the right panel is ≈ 0 at the origin and grows smoothly outward.



Left: contours of the true f (black) and of $Q = y + xy$ (red dashed) almost coincide near the origin. Right: $|f - Q|$ is essentially zero at the centre and grows like the cube of the distance.

Alternate / quicker solution

Multiply the one-variable series and keep total degree ≤ 2:

$$e^x \sin y = \left(1 + x + \tfrac{x^2}{2} + \cdots\right)\left(y - \tfrac{y^3}{6} + \cdots\right) = y + xy + (\text{degree } \ge 3).$$

So $Q = y + xy$ falls out with no Hessian computation. Composing/multiplying standard series and truncating is usually the fastest route to a multivariable Taylor polynomial.

Intuition — why it matters

Near a point a curved surface looks like a quadratic, and the Hessian stores its curvature. Here the pure x- and y-curvatures vanish ($f_{xx} = f_{yy} = 0$); all the second-order behaviour lives in the cross term xy, and $H = \left[\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right]$ has eigenvalues ± 1 — a perfect little saddle.

Problem A3 · Taylor approximation of a value

Let $f(x) = \sqrt{x}$, expanded about the point $a = 4$.

(a) Find the first-order (linear) polynomial $P_1(x)$ and the second-order (quadratic) polynomial $P_2(x)$ of f about $x = 4$.

(b) Use each to estimate $\sqrt{4.2}$.

(c) Using the figure, explain why the quadratic estimate is the better one.

Solution

(a) Derivatives of $f(x) = x^{1/2}$ at $a = 4$:

$$f(4) = 2, \qquad f'(x) = \tfrac{1}{2}x^{-1/2} \Rightarrow f'(4) = \tfrac{1}{4}, \qquad f''(x) = -\tfrac{1}{4}x^{-3/2} \Rightarrow f''(4) = -\tfrac{1}{32}.$$

Hence, about $x = 4$ (using $\dfrac{f''(4)}{2!} = \dfrac{-1/32}{2} = -\dfrac{1}{64}$),

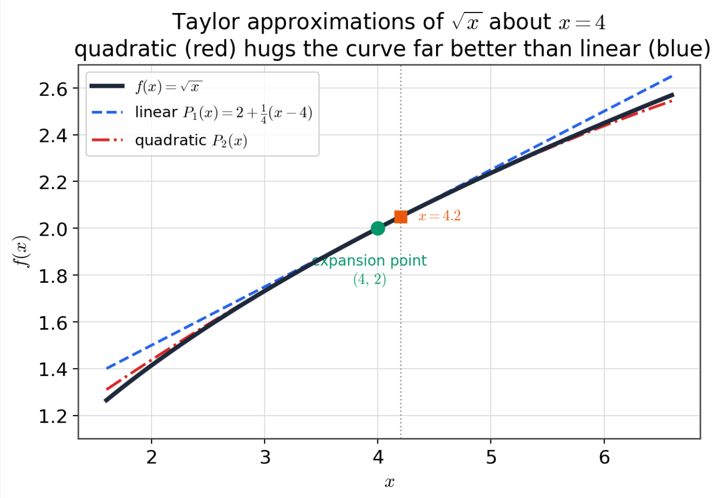
$$P_1(x) = 2 + \tfrac{1}{4}(x-4), \qquad P_2(x) = 2 + \tfrac{1}{4}(x-4) - \tfrac{1}{64}(x-4)^2.$$

(b) At $x = 4.2$ the increment is $x - 4 = 0.2$:

$$P_1(4.2) = 2 + \tfrac{1}{4}(0.2) = 2.05, \qquad P_2(4.2) = 2.05 - \tfrac{1}{64}(0.2)^2 = 2.05 - 0.000625 = 2.049375.$$

(For reference $\sqrt{4.2} = 2.04939\ldots$; the quadratic value 2.049375 matches it far more closely than the linear value 2.05.)

(c) The linear polynomial is the tangent line at $x = 4$: it matches the value and slope of $\sqrt{x}$ there but ignores the downward curvature ($f'' < 0$), so it rides *above* the curve and drifts off. The quadratic adds the $-\frac{1}{64}(x-4)^2$ term, bending the approximation to follow that curvature — in the figure the red curve stays glued to $\sqrt{x}$ over a much wider range than the blue line.



$\sqrt{x}$ with its linear and quadratic Taylor approximations about $x = 4$. The quadratic (red) hugs the curve well beyond the linear (blue) tangent.

Alternate / quicker solution

Factor out the centre and use the binomial series: $\sqrt{4+h} = 2\sqrt{1+\frac{h}{4}} = 2\left(1+\frac{1}{2}\cdot\frac{h}{4}-\frac{1}{8}\cdot\frac{h^2}{16}+\cdots\right) = 2+\frac{h}{4}-\frac{h^2}{64}+\cdots$, with $h = x-4$. This reproduces P_1 and P_2 directly from the known expansion $(1+t)^{1/2} = 1+\frac{1}{2}t-\frac{1}{8}t^2+\cdots$.

Intuition — why it matters

Keeping one Taylor term gives the tangent-line (linearisation) estimate; keeping two adds curvature. Each extra term captures the next derivative, so the approximation improves order by order — the same idea that lets optimisation methods replace a messy loss by a simple quadratic model near a point.

Part B — Maxima, Minima & the Hessian

Problem B1 · Finding and classifying critical points

Let $f(x,y) = x^3 - 3xy + 3y^2$.

(a) Find all critical points.

(b) Form the Hessian and classify each point with the second-derivative test.

(c) State the local-minimum value.

Solution

(a) $\nabla f = \left(3x^2-3y,\ -3x+6y\right)$. From the second equation $x = 2y$; substitute into the first:

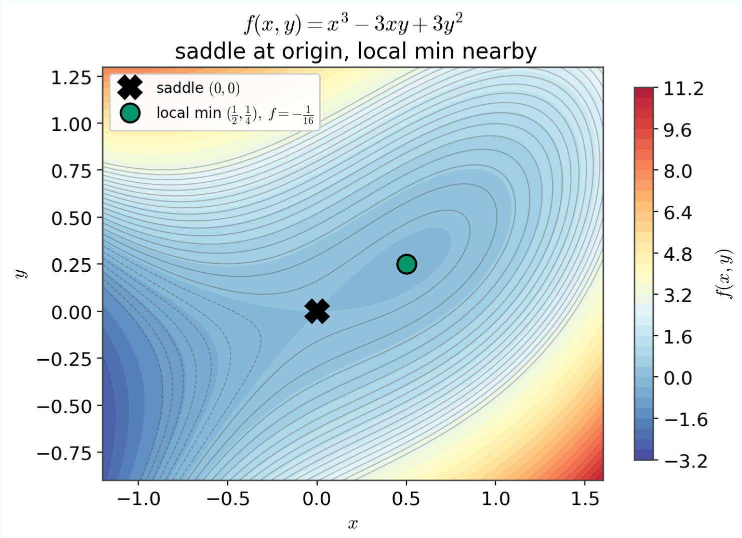
$$3(2y)^2 - 3y = 12y^2 - 3y = 3y(4y-1) = 0 \ \Rightarrow\ y = 0 \text{ or } y = \tfrac{1}{4}.$$

$y = 0 \Rightarrow x = 0$ and $y = \frac{1}{4} \Rightarrow x = \frac{1}{2}$. Critical points: $(0,0)$ and $\left(\frac{1}{2},\frac{1}{4}\right)$.

(b) The Hessian and its discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ are

$$H = \begin{bmatrix} f_{xx} & f_{xy} \\ f_{xy} & f_{yy} \end{bmatrix} = \begin{bmatrix} 6x & -3 \\ -3 & 6 \end{bmatrix}, \qquad D = 36x - 9.$$

At $(0,0)$*:* $D = -9 < 0 \Rightarrow$ **saddle point**.
At $(\tfrac{1}{2}, \tfrac{1}{4})$*:* $D = 36(\tfrac{1}{2}) - 9 = 9 > 0$ and $f_{xx} = 3 > 0 \Rightarrow$ **local minimum**.
(c) $f(\tfrac{1}{2}, \tfrac{1}{4}) = \tfrac{1}{8} - 3(\tfrac{1}{2})(\tfrac{1}{4}) + 3(\tfrac{1}{4})^2 = \tfrac{1}{8} - \tfrac{3}{8} + \tfrac{3}{16} = -\tfrac{1}{16}$.



Contours of f: the $\times$ marks the saddle at the origin, the dot the nearby local minimum $(\tfrac{1}{2}, \tfrac{1}{4})$ with $f = -\tfrac{1}{16}$.

Alternate / quicker solution

Use eigenvalues rather than the discriminant. At $(\tfrac{1}{2}, \tfrac{1}{4})$, $H = \left[\begin{smallmatrix} 3 & -3 \\ -3 & 6 \end{smallmatrix}\right]$ has trace 9 and determinant 9, so $\lambda = \frac{9 \pm \sqrt{81-36}}{2} = \frac{9 \pm \sqrt{45}}{2}$, both positive $\Rightarrow$ positive-definite $\Rightarrow$ minimum. At $(0,0)$, $\det H = -9 < 0$, so the eigenvalues have opposite signs (indefinite) $\Rightarrow$ saddle. (For a 2×2 symmetric matrix $\det H = \lambda_1 \lambda_2$, which is exactly why $D < 0$ forces a saddle.)

Intuition — why it matters

The discriminant D *is* $\det H$. The lecture showed the nature of a critical point is governed by the quadratic form $Q = h^2 f_{xx} + 2hk f_{xy} + k^2 f_{yy}$: $D > 0$ with $f_{xx} > 0$ makes it a positive-definite "bowl" (min); $D < 0$ makes it an indefinite "saddle."

Problem B2 · A positive-definite Hessian (Sylvester & eigenvalues)

A model is trained by minimising the quadratic loss

$$f(x,y) = 2x^2 + 2xy + 3y^2 - 4x - 2y.$$

(a) Find the unique critical point.

(b) Show two ways — Sylvester's criterion and the eigenvalues — that the Hessian is positive-definite, and conclude the point is the *global* minimum.

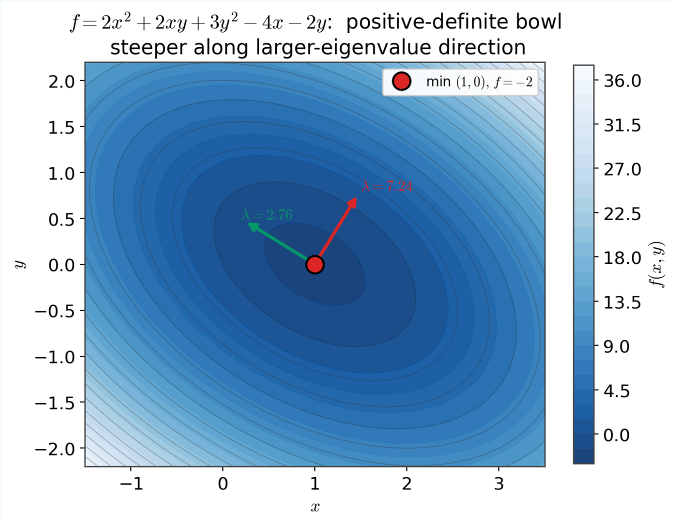
(c) State the minimum value.

Solution

(a) $\nabla f = (4x + 2y - 4,\ 2x + 6y - 2) = \mathbf{0}$, i.e. $2x + y = 2$ and $x + 3y = 1$. From the first $y = 2 - 2x$; substitute: $x + 3(2 - 2x) = 1 \Rightarrow -5x = -5 \Rightarrow x = 1,\ y = 0$. Critical point $(1, 0)$.
(b) Since f is quadratic the Hessian is constant:

$$H = \begin{bmatrix} 4 & 2 \\ 2 & 6 \end{bmatrix}.$$

Sylvester: the leading principal minors are $4 > 0$ and $\det H = 4 \cdot 6 - 2 \cdot 2 = 20 > 0$, so $H \succ 0$. *Eigenvalues:* trace 10, determinant 20, so $\lambda = \frac{10 \pm \sqrt{100-80}}{2} = 5 \pm \sqrt{5} \approx \{2.76,\ 7.24\}$, both positive $\Rightarrow H \succ 0$. Because $H \succ 0$ everywhere, f is strictly convex, so its single stationary point is the global minimum.
(c) $f(1, 0) = 2 - 4 = -2$.



The convex bowl with the Hessian's eigenvectors drawn at the minimum: steepest ascent along the $\lambda \approx 7.24$ direction, gentlest along $\lambda \approx 2.76$.

Alternate / quicker solution

Complete the square. Shift to the critical point with $x = 1 + u,\ y = v$:

$$f = 2(1+u)^2 + 2(1+u)v + 3v^2 - 4(1+u) - 2v = -2 + \left(2u^2 + 2uv + 3v^2\right).$$

Then $2u^2 + 2uv + 3v^2 = 2\left(u + \frac{v}{2}\right)^2 + \frac{5}{2}v^2 \geq 0$, with equality only at $u = v = 0$. Hence $f(x, y) \geq -2$ everywhere, and the global minimum -2 is attained at $(1, 0)$. One manoeuvre proves definiteness *and* pins the value.

Intuition — why it matters

Positive-definite Hessian = symmetric matrix with all-positive eigenvalues = convex bowl, exactly as in the lecture. The condition number $7.24/2.76 \approx 2.6$ measures how elongated the contours are — and it is precisely what governs how quickly gradient descent spirals into the minimum.

Problem B3 · When the second-derivative test fails

Consider $g(x, y) = x^4 + y^4$ and $h(x, y) = x^4 - y^4$.

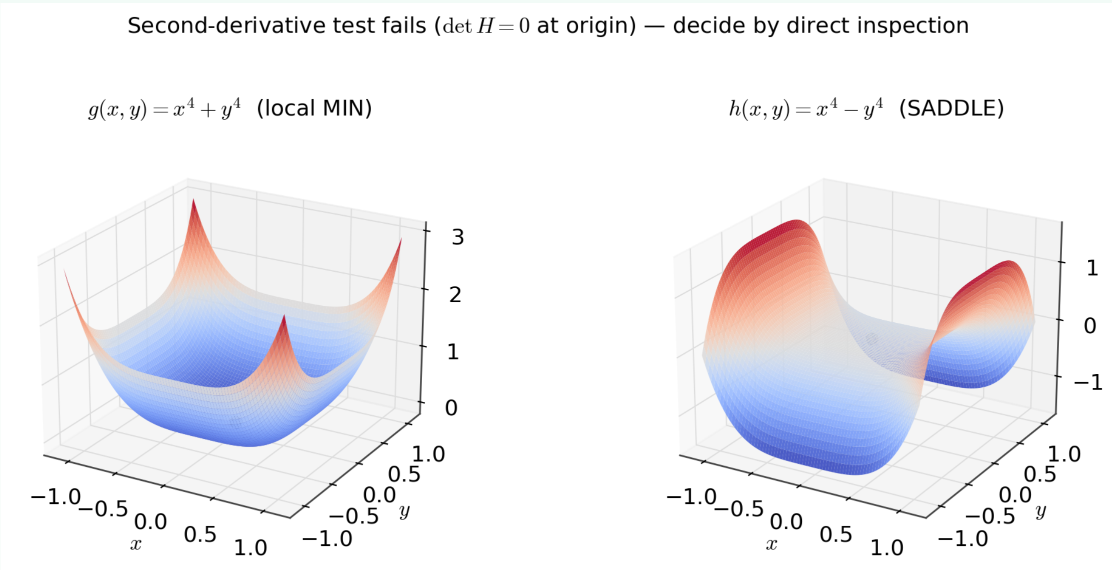
(a) Show that for both, the origin is a critical point and the Hessian there is the *zero* matrix, so the second-derivative test is inconclusive.

(b) Determine the true nature of the origin for each by direct argument.

(c) In one line, say why the Hessian could not distinguish them.

Solution

(a) $\nabla g = (4x^3, 4y^3) = \mathbf{0}$ at the origin, and $H_g = \begin{bmatrix} 12x^2 & 0 \\ 0 & 12y^2 \end{bmatrix} = \left[\begin{smallmatrix} 0 & 0 \\ 0 & 0 \end{smallmatrix}\right]$ there. Identically $\nabla h = (4x^3, -4y^3) = \mathbf{0}$ and $H_h = \begin{bmatrix} 12x^2 & 0 \\ 0 & -12y^2 \end{bmatrix} = \mathbf{0}$ at the origin. So $D = \det H = 0$ for both — the test says nothing (the lecture's "$f_{xx}f_{yy} - f_{xy}^2 = 0 \Rightarrow$ another test is needed").
(b) *For g:* $g(x,y) = x^4 + y^4 \geq 0$ for all (x,y), with equality only at the origin $\Rightarrow$ strict local (indeed global) **minimum**.
For h: along the x-axis $h(t,0) = t^4 > 0$, but along the y-axis $h(0,t) = -t^4 < 0$. A point that is a minimum in one direction and a maximum in another is a **saddle**.
(c) The Hessian captures only second-order behaviour; here the leading behaviour is *quartic*, which a zero Hessian misses entirely.



Both have a flat (zero) Hessian at the origin, yet $g = x^4 + y^4$ is a bowl (minimum) while $h = x^4 - y^4$ is a saddle — the difference is fourth-order.

Alternate / quicker solution

Slice along a ray $(x,y) = t(\cos\theta, \sin\theta)$. Then $g = t^4(\cos^4\theta + \sin^4\theta) \geq 0$ for every $\theta \Rightarrow$ minimum. But $h = t^4(\cos^4\theta - \sin^4\theta) = t^4 \cos 2\theta$ changes sign with θ (positive for $|\theta| < 45°$, negative beyond) $\Rightarrow$ saddle. Directional curvature that changes sign with direction is the signature of a saddle.

Intuition — why it matters

A singular (zero) Hessian is the boundary case the lecture flags, and it is exactly what makes flat plateaus in neural-network loss surfaces hard to analyse from curvature alone — you must look at higher-order terms or the global shape.

Part C — Backpropagation & Automatic Differentiation

Problem C1 · Chain rule through a linear layer and a squared loss

A one-layer network computes $\mathbf{z} = W\mathbf{x}$ and the loss $g(\mathbf{z}) = \frac{1}{2}\|\mathbf{z} - \mathbf{t}\|^2$ against a target $\mathbf{t}$. By the chain rule $\nabla_{\mathbf{x}}\, g = J^\top \nabla_{\mathbf{z}}\, g$ with $J = \partial\mathbf{z}/\partial\mathbf{x}$. Take

$$W = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}, \qquad \mathbf{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \mathbf{t} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}.$$

(a) Compute $\nabla_{\mathbf{z}}\, g$ (state its dimension) and $J = \partial\mathbf{z}/\partial\mathbf{x}$ (state its dimension).

(b) Combine them to get $\nabla_{\mathbf{x}}\, g$, and give the loss value.

Solution

Forward pass. $\mathbf{z} = W\mathbf{x} = (1\cdot 1 + 2\cdot 1,\ 0\cdot 1 + 1\cdot 1,\ -1\cdot 1 + 1\cdot 1)^\top = (3, 1, 0)^\top$. Residual $\mathbf{r} = \mathbf{z} - \mathbf{t} = (2, 1, -2)^\top$, and $g = \frac{1}{2}(2^2 + 1^2 + (-2)^2) = \frac{9}{2} = 4.5$.
(a) For $g = \frac{1}{2}\|\mathbf{z} - \mathbf{t}\|^2$, $\nabla_{\mathbf{z}}\, g = \mathbf{z} - \mathbf{t} = (2, 1, -2)^\top$, dimension 3×1 (one entry per output). The Jacobian of a linear map is the matrix itself, $J = \partial(W\mathbf{x})/\partial\mathbf{x} = W$, dimension 3×2 (outputs $\times$ inputs).
(b) Therefore

$$\nabla_{\mathbf{x}}\, g = J^\top \nabla_{\mathbf{z}}\, g = W^\top \mathbf{r} = \begin{bmatrix} 1 & 0 & -1 \\ 2 & 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 1\cdot 2 + 0\cdot 1 + (-1)(-2) \\ 2\cdot 2 + 1\cdot 1 + 1\cdot(-2) \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}.$$

So $\nabla_{\mathbf{x}}\, g = (4, 3)^\top$, dimension 2×1 — the shape of $\mathbf{x}$, as it must be. *Dimension check:* $\nabla_{\mathbf{z}} g$ is 3×1, $J^\top$ is 2×3, product 2×1. ✓

Alternate / quicker solution

Substitute first: $g(\mathbf{x}) = \frac{1}{2}\|W\mathbf{x} - \mathbf{t}\|^2$ has gradient $W^\top(W\mathbf{x} - \mathbf{t}) = W^\top\mathbf{r}$, the same result. (Expanding $g = \frac{1}{2}\mathbf{x}^\top W^\top W\mathbf{x} - \mathbf{t}^\top W\mathbf{x} + \frac{1}{2}\|\mathbf{t}\|^2$ and differentiating term by term gives $W^\top W\mathbf{x} - W^\top\mathbf{t} = W^\top\mathbf{r}$ as well.)

Intuition — why it matters

This is one backward step of backprop: the upstream gradient $\nabla_{\mathbf{z}} g = \mathbf{r}$ arrives at the layer's output and is passed to its input by left-multiplying with the transpose of the local Jacobian. For a linear layer that Jacobian is just W, so "backprop through a linear layer" = "multiply the incoming gradient by $W^\top$."

Problem C2 · Backpropagation through a single nonlinear neuron

A neuron computes $z = wx + b$, $a = \tanh(z)$, and loss $L = \frac{1}{2}(a - y)^2$. Using $\tanh'(z) = 1 - \tanh^2(z)$ and the values $w = 0.5,\ x = 1,\ b = 0,\ y = 1$:

(a) Do the forward pass (z, a, L).

(b) Backpropagate to obtain $\dfrac{\partial L}{\partial w}, \dfrac{\partial L}{\partial b}, \dfrac{\partial L}{\partial x}$.

Solution

(a) Forward. $z = 0.5 \cdot 1 + 0 = 0.5$; $a = \tanh(0.5) = 0.4621$; $L = \frac{1}{2}(0.4621 - 1)^2 = \frac{1}{2}(-0.5379)^2 = 0.1447$.

(b) Backward along $z \to a \to L$. First the shared "local error"

$$\frac{\partial L}{\partial a} = a - y = -0.5379, \qquad \frac{da}{dz} = 1 - a^2 = 1 - 0.2136 = 0.7864,$$

$$\Rightarrow \frac{\partial L}{\partial z} = \frac{\partial L}{\partial a}\frac{da}{dz} = (-0.5379)(0.7864) = -0.4230.$$

Since $z = wx + b$ gives $\partial z/\partial w = x = 1$, $\partial z/\partial b = 1$, $\partial z/\partial x = w = 0.5$,

$$\frac{\partial L}{\partial w} = \frac{\partial L}{\partial z}x = -0.4230, \qquad \frac{\partial L}{\partial b} = \frac{\partial L}{\partial z} = -0.4230, \qquad \frac{\partial L}{\partial x} = \frac{\partial L}{\partial z}w = -0.2115.$$

Alternate / quicker solution

Write $L(w) = \frac{1}{2}\big(\tanh(wx+b) - y\big)^2$ and differentiate directly: $\dfrac{\partial L}{\partial w} = (a-y)(1-a^2)\,x$, and likewise with factor 1 for b and w for x. Plugging in reproduces $(-0.4230, -0.4230, -0.2115)$. Backprop is exactly this chain rule, organised so that the shared factor $\partial L/\partial z = (a-y)(1-a^2)$ is computed *once* and reused.

Intuition — why it matters

That shared factor $\partial L/\partial z$ (the "delta" δ) is computed once and then handed to every parameter feeding z. Computing δ once and multiplying by each input is why one backward pass costs about the same as one forward pass — not proportional to the number of parameters.

Problem C3 · The multivariable chain rule

A scalar function is built from two intermediate variables,

$$f = u\,v, \qquad u = x^2 + y, \qquad v = x - y.$$

(a) Write $\partial f/\partial u$, $\partial f/\partial v$, and the four partials of u, v with respect to x and y.

(b) Using the chain rule, assemble $\partial f/\partial x$ and $\partial f/\partial y$, and evaluate them at $(x, y) = (1, 1)$.

Solution

(a) Outer derivatives: $\dfrac{\partial f}{\partial u} = v$ and $\dfrac{\partial f}{\partial v} = u$. Inner derivatives:

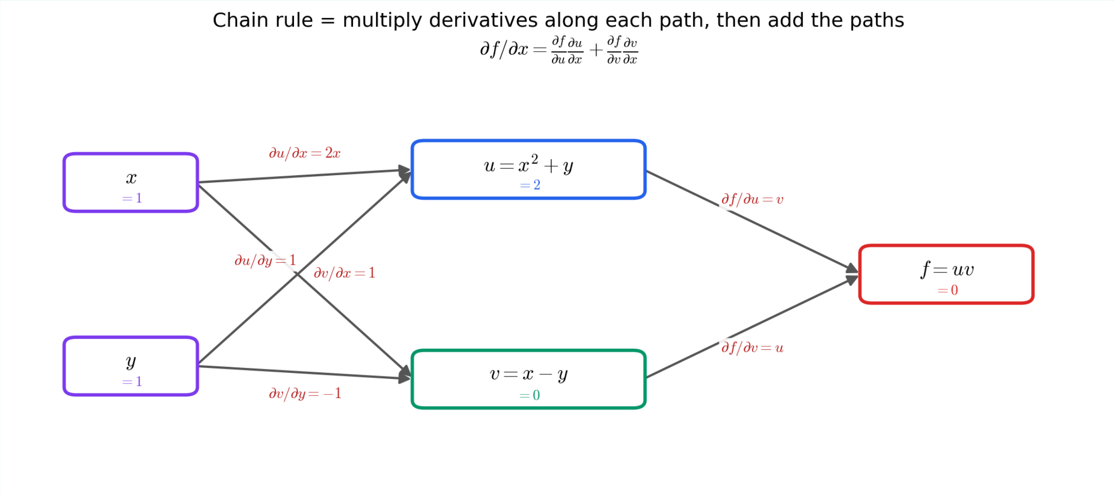
$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial u}{\partial y} = 1, \qquad \frac{\partial v}{\partial x} = 1, \quad \frac{\partial v}{\partial y} = -1.$$

(b) The chain rule sums over both intermediate variables (the two paths from an input to f):

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial x} = v(2x) + u(1) = 2x(x-y) + (x^2+y) = 3x^2 - 2xy + y,$$
$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial u}\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\frac{\partial v}{\partial y} = v(1) + u(-1) = (x-y) - (x^2+y) = -x^2 + x - 2y.$$

At $(x, y) = (1, 1)$ (where $u = 2,\ v = 0$):

$$\left.\frac{\partial f}{\partial x}\right|_{(1,1)} = 3 - 2 + 1 = 2, \qquad \left.\frac{\partial f}{\partial y}\right|_{(1,1)} = -1 + 1 - 2 = -2.$$



The chain rule as paths: each arrow carries a local derivative, and $\partial f/\partial x$ is the sum over the two routes $x \to u \to f$ and $x \to v \to f$.

Alternate / quicker solution

Substitute first, then differentiate: $f = (x^2 + y)(x - y) = x^3 - x^2y + xy - y^2$, so $\partial f/\partial x = 3x^2 - 2xy + y$ and $\partial f/\partial y = -x^2 + x - 2y$ — identical to the chain-rule result, giving 2 and -2 at $(1, 1)$. Expanding first is quick here, but the chain rule is what scales to deep compositions you cannot expand by hand.

Intuition — why it matters

This is the engine inside backpropagation: to differentiate a composition you multiply the local derivatives along each path from input to output, then add the paths. When an input feeds several intermediates (here x feeds both u and v), its derivative is the *sum* of the per-path contributions.